Software Requirement and Specification

for

Sky-FLOW (Real-Time Air-Traffic Simulation System)

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Contents

1 Introduction

1.1 Purpose of the Document

The purpose of this document is to formally capture and document the requirements obtained from the client interview process for the Real-Time Air-Traffic Simulation System. This SRS serves as a reference for system design, implementation, and evaluation.

1.2 Scope of the Project

The Real-Time Air-Traffic Simulation System is a web-based application designed to visually simulate aircraft movement in real time. The system focuses on aircraft-level simulation and visualization and explicitly excludes non-simulation features such as ticket booking.

2 Client Interview Summary

The Interview Invitation was sent on Sunday, 18-01-2026 at 9:00 PM. The interview was held on the same day at 10:00 PM. The interview was 15 minutes long on an online Zoom meeting.

3 Client Interview Questionnaire

3.1 General Information

Question	**Client Response**
What is the primary objective of the air-traffic simulation system?	For Visualization of Flight Traffic
Is the system intended for?	General users
Should the system represent?	Real-world air traffic simplified simulation

3.2 Target Audience & Usage

Question	Client Response
What is the experience level of the intended users?	Beginner to Intermediate
What is the target age group?	Adults (18 to 40)
Will the system be accessed primarily on?	Desktop/laptop and mobile both
Approximately how many users may access the system simultaneously?	1000-2000

3.3 Simulation Scope & Coverage

Question	Client Response
What geographical scope should the simulation cover?	Global airspace (changes made in the constraints)
Which types of flights should be included?	Commercial and Cargo Both (can also add military) both aeroplane and helicopter
Should real airline names and identifiers be used, or generic placeholders?	Yes
Should aircraft follow realistic flight paths or simplified routes?	Simplified route would be enough
Should weather conditions affect flight behavior?	It's up to the flight (we don't decide if the weather affects the flight or not)

3.4 Core Simulation Features

Question	Client Response
Should aircraft movement be displayed in real time on a map?	Yes, its route after you click on it.
Should users be able to zoom and pan the map?	Zoom and pan the Map
What flight information should be visible?	Flight ID, Origin Destination, flight occupancy
Should the simulation include takeoff or landing?	No takeoff or Landing, just En-route Movement is required
Do you want collision detection or alert mechanisms?	No, we don't need it

3.5 Visualization & UI/UX Preferences

Client's suggested inspired website: FlightRadar24

Question	**Client Response**
What overall design style do you prefer?	Minimalistic
What type of map view should be used?	Satellite View would be preferred
How should aircraft be represented on the map?	Size based on aircraft type
Should aircraft details be?	Flight ID should be visible on hover and remaining details visible only after clicking
Do you have any color preferences for aircraft icons?	Color Coded will be nice

3.6 User Interaction & Control

Question	**Client Response**
Should users be able to start/pause the simulation and control simulation speed?	Yes, they can
Should there be different user roles?	No, every user has same ability to control or speed the simulation
Should users be able to manually add or remove flights?	If you click on the flight then only that flight will be visible, remaining will disappear (till you unclick the flight)

3.7 Scope Control & Constraints

Question	**Client Response**
Should the system include ticket booking functionality?	No, It's just Simulation
Should passenger-level data be shown, or only aircraft-level information?	Only aircraft-level information
Should the system display analytics such as active flights or average altitude/speed?	Yes
Are there any features that should be explicitly excluded?	No place for advertisements
What is the expected project deadline?	April 15, 2026

4 Functional Requirements

4.1 Use-Case Diagram

The following use-case diagram illustrates the primary interactions between users and the Real-Time Air-Traffic Simulation System. It represents the functional scope of the system derived from the requirement gathering phase.

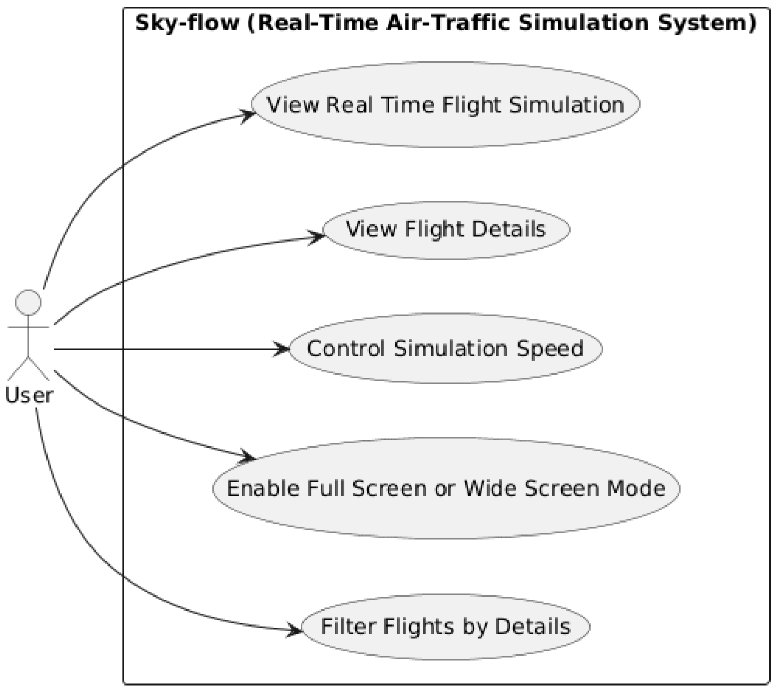

Figure 1: Use-Case Diagram for Real-Time Air-Traffic Simulation System

Requirement Name	Description
Retrieve Air-Traffic Data	Retrieve real-time aircraft data from OpenSky Network API within defined geographical scope.
Geographical Scope Limitation	Restrict data visualization to Indian airspace to comply with API usage constraints.
Display Aircraft on Map	Display active aircraft positions on interactive map using visual markers.
Real-Time Aircraft Movement	Update aircraft positions dynamically to simulate real-time flight movement.
Aircraft Information Display	Allow users to view aircraft details (flight ID, altitude, speed, origin-destination) on interaction.
Map Interaction Controls	Enable users to zoom, pan, and navigate the map within supported airspace.
Simulation Control	Provide controls to start, pause, and resume air-traffic simulation.
Simulation Speed Adjustment	Allow users to adjust simulation speed for different time scales.
Visualization Customization	Represent aircraft using configurable attributes (color, size, icons) based on predefined rules.

5 Non-Functional Requirements

Performance Requirements

- The system shall update aircraft positions at a controlled refresh interval to ensure smooth visualization without exceeding API request limits.
- The system shall load the initial simulation view within an acceptable time under normal network conditions.
- The system shall limit unnecessary API calls to comply with third-party data service constraints.

6 Constraints and Assumptions

6.1 Constraints

- The system uses third-party aviation data of free-tier basis services. This is because of the scarcity of free API tokens and request. The amount of real time calls of data depends on quotas.
- The OpenSky Network API has a maximum of about eight hundred API calls in a day as a free usage policy. This limitation limits the number of real-time data requests and their quantities.

- Due to the limitations on API described above, the scope of simulation is constrained geographically. Air traffic will be simulated by the system primarily in the Indian airspace as opposed to international coverage.

6.2 Assumptions

The design and development is done based on the following assumptions of the system:

- OpenSky Network API will be open and accessible during the development and demonstration.
- Reliability of aircraft information requires the timeliness and accuracy of aircraft information that is presented through the external API and is deemed to be credible to be educative and graphic.
- System users will enjoy a stable internet connection to retrieve real time or nearly real time simulation data.
- The system is also designed to be used in academic and educational purposes, and small delay or approximation of real-time information is tolerable.

7 Conclusion

The Real-Time air-traffic Simulation requirement gathering phase. System has been successfully done under organized client. interviews and analysis. The requirements are documented giving a clear picture. and clear basis to the structure and execution of the. system.

But as the project moves on to development, small revisions or improvements to the requirements can be required. implement real life application and enhance efficiency of the system. These changes will be handled with caution to make them in line with the. overall project objectives.

The needs recorded in this report will be the ones that are supposed to guide the. Effective implementation of the system, but at the same time, making it possible to control. improvements in new development stages.